

READY TO FLY An Indian Army soldier handles a drone during Exercise Topchi in Nashik, Jan. 2026

PTI

is not an attempt to replicate the Shahed but is India's answer to what comes after it. "We are building a mass-deployable strike platform designed for swarm operations—100 to 300 drones launched within a short operational window, overwhelming enemy air defences through sheer saturation. The design is modular, built for rapid manufacturing and mission flexibility across strike, decoy and EW roles," he says. Noting that the Shahed changed the calculus of modern conflict by proving that volume beats precision at scale, Sanghapriya adds, "KAL is designed to give India that same asymmetric edge on Indian terms."

Mass Expendable Fleets

Sameer Joshi, CEO of NRT, which is developing the Sheshnag-150 drone designed to strike targets over 1,000 km away, says the Indian military is prioritising thousands of low-cost, attritable UAVs for saturation and deep-strike operations. He notes that the MoD is integrating dedicated drone platoons and counter-UAS (Unmanned Aerial Systems) units across infantry and mechanised units. While high-end platforms remain for reconnaissance/ surveillance, mass expendable fleets now form the doctrinal core for asymmetric advantage along the Line of Control (LoC) and the Line of Actual Control (LAC).

"India's UAV ecosystem has been crippled for years by fragmented procurement, the toxic lowest-bidder trap and a dangerously thin domestic component base. Operation Sindoor laid bare these weaknesses and created a now-or-never moment for reform," Joshi says. He adds that the industry needs multi-year committed orders, turbocharged PLI (production linked incentive) schemes for true indigenisation and procurement models that reward deep domestic system-level innovation, not just the lowest price. He believes that success demands nothing less than full doctrinal integration of UAVs across services, tight public-private R&D fusion and the mass fielding of thousands of low-cost platforms by 2030. Today, 50–60 per cent of mission-critical subsystems remain imported, leaving sovereign capability exposed. "The proposed PLI for FY 2026–27, reinforced by the Centre's Electronics Components Manufacturing Scheme, must drive import dependence below 40 per cent by this decade's end. Anything more jeopardises India's ability to sustain drone warfare on our terms," says Joshi.



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Sai Pattabiram, co-founder of Chennai-based Zuppa Geo Navigation Technologies, a drone firm specialising in software, says ongoing global conflicts demand a paradigm shift—performance metrics now focus on the asymmetric cost impact of equipment in both attack and defence roles. Consequently, cost-effective equipment must be produced, procured and deployed in millions. "Rapid adaptability in weeks across the entire inventory to evolving threats and countermeasures is crucial. The MoD must lead this evolution," he says.

A top military official says the biggest constraint is scale. Over 600 Indian firms manufacture drones and UAV equipment but the expanding drone ecosystem still lacks industrial-level production capacity of cheap drones in large numbers.

"The future of warfare will not be decided solely by the most advanced technology but by the ability of drones to endure in a jammed, silent battlespace—and to fight back with systems that are cheap, numerous, intelligent and impossible to ignore," he adds.

The challenge is thus twofold—achieving fully indigenous, state-of-the-art technology and ensuring its rapid manufacture. Finally, India is gearing up to meet it. ■